Software Engineering: Algorithms and Data Structures
Example 2
MARK: 63%

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//============================================================================
===
// Name        : testB.cpp
// Author      :
// Version     :
// Copyright   : Your copyright notice
// Description : Hello World in C++, Ansi-style
//============================================================================
===

#include <iostream>
#include <fstream>
#include <sstream>
#include <string>

using namespace std;
typedef string T; //T == dataType to be used
struct node{
                T val;
                node* next;
        };

typedef node* listPtr;


void insmlistend(T el, listPtr& l){
                if(l == NULL){
                        listPtr tmpPtr = new node;
                        tmpPtr -> val = el;
                        tmpPtr -> next = NULL;
                        l = tmpPtr;
                }
                else {
                        insmlistend(el, l->next);
                }
}

void printmlist(stringstream& _ss, listPtr l){
        if(l != NULL){
                _ss << l->val << endl;
                if (l->next != NULL){
```

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                printmlist(_ss, l->next);
            }

        }
}


bool ismlistsorted(listPtr l){
        if(l != NULL){
            if(l->next == NULL){
                return true;//base case if it was true before it will be true now
            }
            else if (l->val < l->next->val){
                //next is bigger continue search
                return ismlistsorted(l->next);
            }
            else {
                //we have found an unsorted item
                //no need to traverse anymore
                return false;
            }
        }
        else {
            return true;//base case, empty list is assumed sorted.
        }
}

void swapList(){

}

int main() {

        listPtr mainList = NULL;
        listPtr tempPtr;
        T tmp;

        cout << "enter element" << endl; // prints !!!Hello World!!!
        cin >> tmp;

        while(tmp != "stop"){
            tempPtr = mainList;
            insmlistend(tmp, tempPtr);
            if (mainList == NULL){          /// the code here is not clean.
                mainList = tempPtr;
            }
            cin >> tmp;
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        }
        stringstream ss;
        printmlist(ss, mainList); //print list to stringstream
        cout << "the items are: \n" << ss.str() << endl;

        cout << "the items are: " << (ismlistsorted(mainList) ? "sorted" : "not sorted") <<
endl;
        return 0;
}
```